

ABIVisualizer — Algorithm Visualizer

CSCI 22042 Visual Programming

University of Kelaniya · 2024/2025 Semester II

ABIVisualizer is a desktop application developed using C# and WinForms to help students understand how sorting and pathfinding algorithms work through real-time visual animations.

The application provides an interactive and user-friendly environment where users can observe algorithms step-by-step, making it easier to learn concepts such as traversal, comparisons, swapping, shortest-path searching, and heuristic-based searching.

Features - Sorting Visualizer

- Insertion Sort
- Merge Sort
- Quick Sort
- Shell Sort
- Heap Sort
- Real-time sorting animation
- Color-coded visualization
- Live comparison counter
- Start, Pause, Resume, and Reset controls

Features - Pathfinding Visualizer

- Breadth First Search (BFS)
- Dijkstra's Algorithm
- A* Algorithm
- Greedy Best-First Search
- Interactive grid system
- Place Start and End nodes
- Create walls and obstacles
- Animated traversal process
- Final shortest path highlighting

Technologies Used

- C#
- .NET 8
- Windows Forms (WinForms)

- GDI+ Graphics Rendering
- Object-Oriented Programming (OOP)

Requirements

- Visual Studio 2022
- .NET 8 SDK or later
- Windows 10 or Windows 11

Future Improvements

- Additional sorting algorithms
- Maze generation algorithms
- Algorithm comparison mode
- Performance statistics panel
- Theme customization

Run Using Command Line

```
cd ABIVisualizer  
dotnet run
```

Author

Kalavarthanan Jogeswaran
Faculty of Computing and Technology
University of Kelaniya